

Guillermo Mendoza

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Education

Pitzer College, Claremont, CA May 2026
B.A. in Neuroscience with Honors & Data Science GPA: 3.80
Davis Scholar, Pitzer Scholar, Sigma Xi Member
Relevant Coursework: Applications of AI Engineering, Machine Learning, Neural Networks, Data Structures & Algorithms, Statistical Linear Models, Probability, Statistical Inference, Big Data, Intro Computational Neuroscience

Experience

Research Assistant Jan 2024 – May 2026
Beat Lab (Computational Neuroscience Lab) Claremont, CA

- Developed and evaluated machine-learning models for high-dimensional motor-imagery EEG, benchmarking real-valued CSP, complex-valued CSP, and a complex-valued neural network across subjects.
- Built Python pipelines spanning filtering, Hilbert-transform analytic signal construction, whitening, trial segmentation, feature extraction, model training, cross-validation, hyperparameter optimization, and spectral analysis.
- Extended Common Spatial Patterns into the complex domain to preserve phase relationships across channels; complex-valued CSP consistently outperformed real-valued CSP across analyzed subjects and component counts.

Web Developer (Portal & Forms) Jul 2025 – May 2026
Pitzer College Claremont, CA

- Designed and implemented student-facing digital forms and portal components using HTML and JavaScript, improving structured data capture and administrative workflows.
- Translated stakeholder needs into technical requirements and collaborated with campus teams to test workflows, diagnose usability issues, and iteratively refine internal tools.

AI Consultant – Airis Garden Practicum Jan 2026 – May 2026
Airis Garden Remote

- Advised a Scandinavian plant-tech startup on LLM-native marketing, AI discoverability, conversion, and chatbot deployment while translating ambiguous client needs into technical and business recommendations.
- Analyzed advertising, sales, and market signals and built an AI “Plant Coach” chatbot concept; communicated findings through weekly executive memos and presentations to the CEO and CMO.

Research Assistant May 2023 – Aug 2025
Sheung Lab (Experimental Biophysics Lab) Claremont, CA

- Modeled emergent dynamics in active-matter systems experimental data using MATLAB.
- Managed research workflows and trained 8 research assistants on standardized protocols and reproducible experimental procedures.

Selected AI & Engineering Projects

Job Application Tracker – Production LLM Automation Sep 2026

- Built a production automation that reads job-application emails, uses Gemini on Vertex AI for structured event extraction, and applies deterministic Python logic for application matching, validation, and Google Sheets updates.
- Deployed the Dockerized workflow as a scheduled Google Cloud Run Job with Firestore persistence, Cloud Scheduler, ambiguous-case review queues, separate read/write OAuth credentials, and structured operational logging.
- Designed fail-closed safeguards, retry handling, configuration validation, and automated tests so uncertain or malformed model output cannot silently trigger unintended persistent writes.

AI Resume Builder – Grounded Document Generation Jul 2026

- Built a Gradio application that extracts job requirements, retrieves supporting evidence from resumes and public GitHub metadata, and generates source-grounded resume and cover-letter recommendations with Gemini.
- Implemented requirement-coverage reporting, evidence tracing, document diffs, unsupported-claim checks, automated tests, and GitHub Actions CI to expose generation decisions and reduce unsupported output.

Los Angeles Parks RAG Guide Jun 2026

- Built a modular RAG application over 664 chunks from 10 community, government, review, and travel sources using source-specific preprocessing, vector retrieval, grounded generation, and a Gradio interface.
- Evaluated retrieval and generated responses across five query categories, reproduced a grounding failure, traced it to weak semantic retrieval, and designed a distance-threshold fallback to block unsupported outputs.

Technical Skills

Programming: Python, JavaScript, TypeScript, SQL, R, MATLAB, C++

AI / Machine Learning: Gemini, Vertex AI, scikit-learn, Keras/TensorFlow, RAG, structured extraction, retrieval evaluation, feature engineering, cross-validation, hyperparameter optimization, model validation, error analysis

Cloud & Data: Google Cloud Run Jobs, Cloud Scheduler, Firestore, PostgreSQL, Docker, REST APIs, OAuth, large-scale data processing

Engineering: Git, GitHub Actions, Linux, pytest, Pydantic, shell scripting, API logging, retry handling, rate limiting